

Motivation

While humans excel at learning new concepts from limited data, deep learning relies on having large annotated datasets. **Prototypical Networks [1]** address this by being a **data-efficient few-shot approach** for unseen classes. It learns an **embedding network** during training that can be **generalized to create prototypes** of new test classes with limited examples for classification.

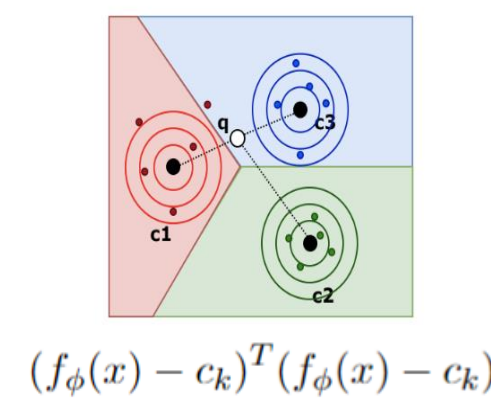
Theory

The probability $p_\phi(y = k|x)$ of query, x belonging to class, k is given by a **softmax** over **negative distances** of embeddings $f(x)$ to **prototypes parameterized from supports**, μ_k .

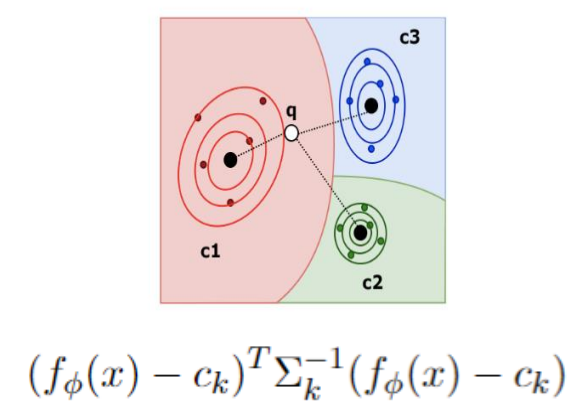
$$p_\phi(y = k|x) = \frac{\exp(-d(f_\phi(x), \mu_k))}{\sum_{k'} \exp(-d(f_\phi(x), \mu_{k'}))}$$

Under this specification, **Bregman Divergences** are distance functions that describe **probabilistic clusters** for classes

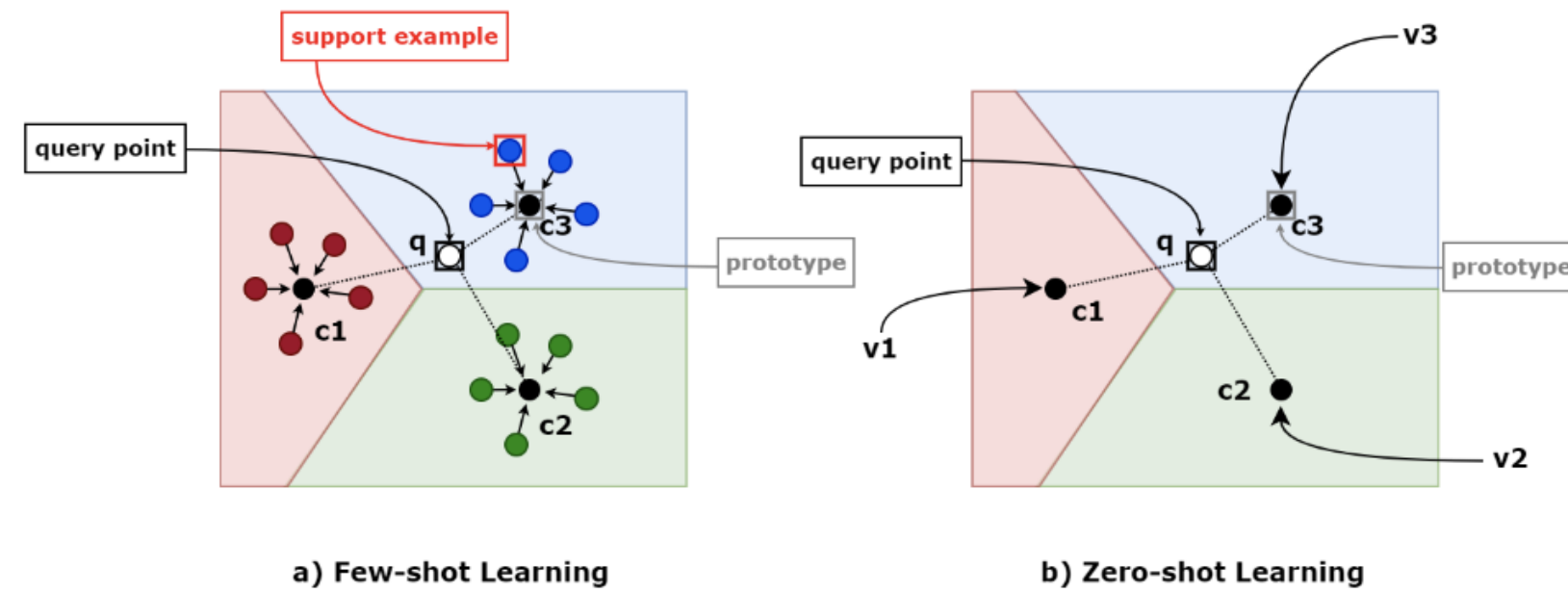
The **Euclidean Distance** describes **Gaussian Clusters** parameterised by a single **mean**, c_k , as the prototype. The MLE is the **empirical mean** of the support. This distance metric assumes a **spherical variance**, resulting in a **linear decision boundary**



The **Mahalanobis Distance** also describes **Gaussian Clusters** but **relaxes the assumption of spherical variances**, requiring an additional **estimate of a class-specific covariance**, Σ_k , to describe a **quadratic decision boundary**



Definitions



Prototypes: Vector representation of class

Support: Examples used to create prototype

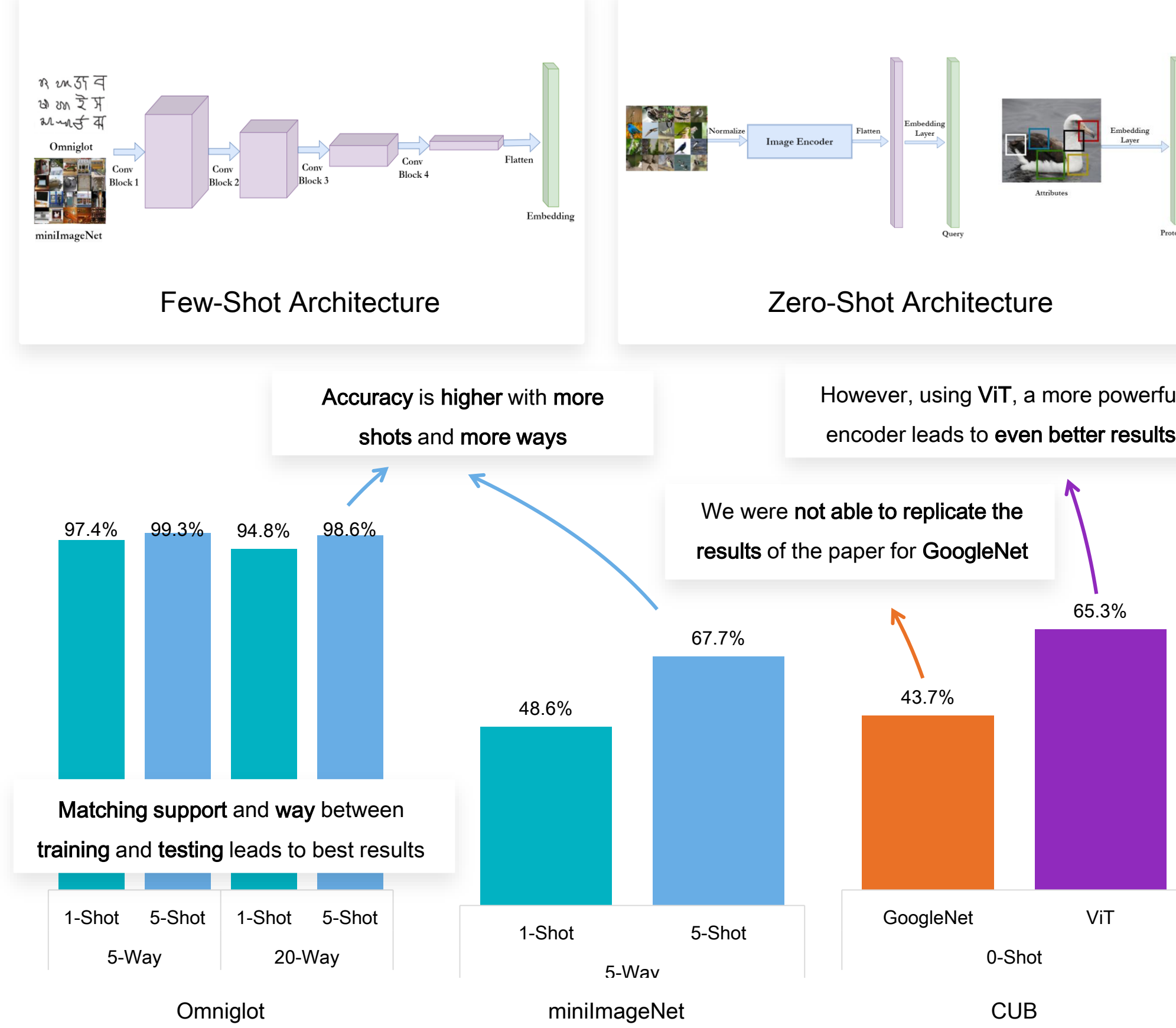
Shot: Number of supports given

Query: Tests used for classification

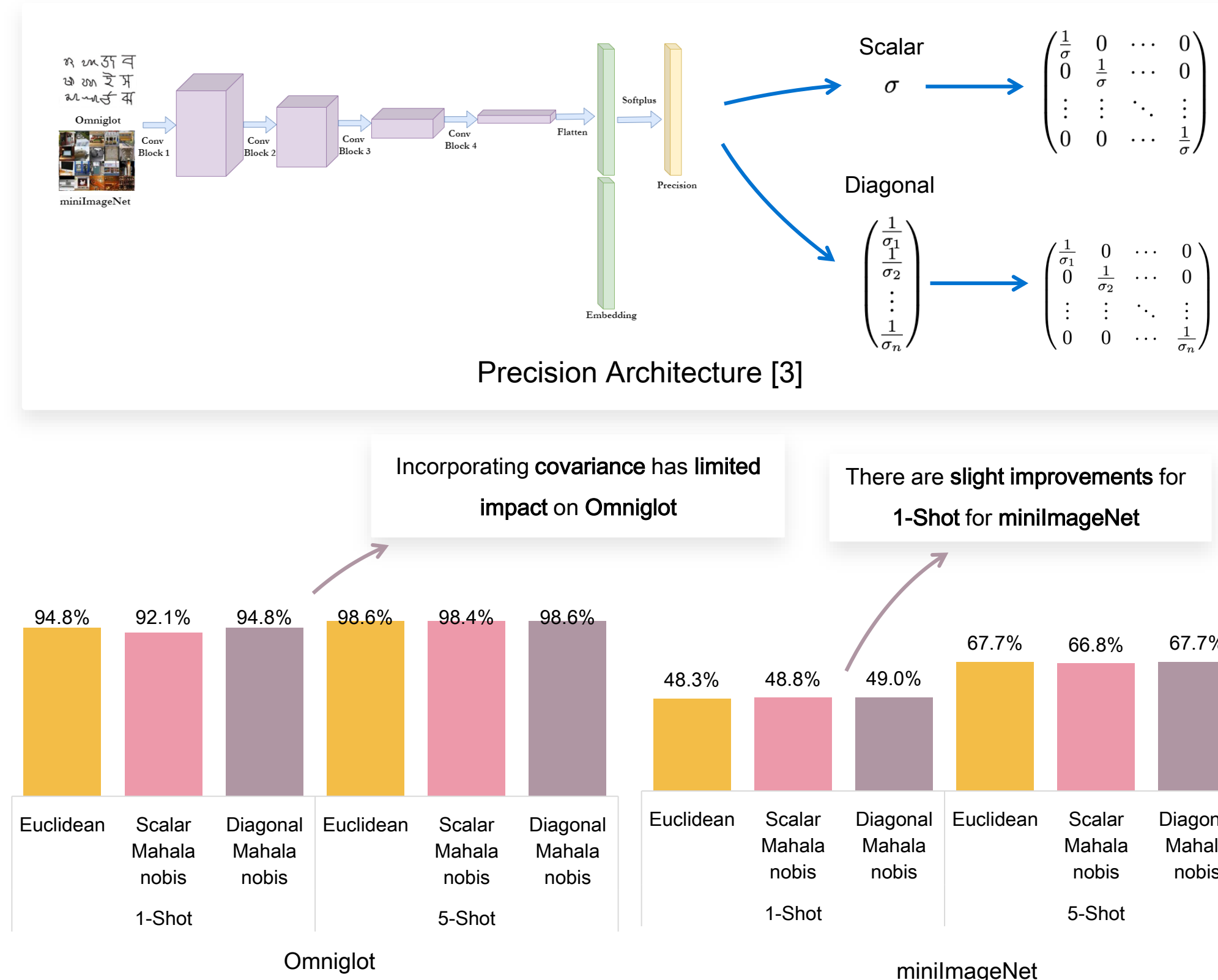
Way: Number of classes to classify against

Distance: Between query and prototypes

Euclidean Distance



Mahalanobis Distance



Prototype Refinement



Takeaways

- Utilizing class-specific covariances through the Mahalanobis distance enhances one-shot classification in more complex datasets
- Refining the predicted covariances using Bregman Soft Clustering boosts both one-shot and five-shot classification on all datasets
- Having a more powerful pre-trained backbones can significantly improve zero-shot classification

Future Work

We can investigate the **robustness of domain shifts** by evaluating performance on **out-of-domain classification** tasks. Additionally, we can modify the architecture to explore **Mahalanobis Distance for zero-shot classification**. We could also add **class-specific adaptive FiLM layers** to the backbone inspired by **CNAPS**.